National University of Computer and Emerging Sciences
CS3009 — Software Engineering | Spring 2026

Deliverable 2

Naawbi

Learning Management System

Team Name	Ibwaan
Team Lead	Ibraheem Farooq
Members	23i-0708 | Ali Aan Khowaja 23i-0816 | Ibraheem Farooq 23i-0819 | Wajih-Ur-Raza Asif
Submission Date	25 March 2026
Repository	github.com/aliaankhowaja/Naawbi

PART A — Project Team Info

Team Updates

No changes to team membership have occurred since Deliverable 1. Ibraheem Farooq continues to serve as Team Lead. All three team members additionally contributed as developers during Sprint 2.

Member	Role(s)	GitHub
Ibraheem Farooq	PM / Scrum Master / Developer	github.com/ibbi1020
Wajih-Ur-Raza Asif	Requirements Architect / Tester / Developer	github.com/wajih-rathore
Ali Aan Khowaja	Developer / UI Designer	github.com/aliaankhowaja

Team Agreement

The team agreement established in Deliverable 1 remains fully active. No revisions to communication methods, meeting schedules, version control practices, work division, submission responsibilities, or contingency planning were required during Sprint 2.

PART B — Sprint Info

1. Sprint 2 Overview

Field	Value
Sprint Duration	1 March 2026 – 25 March 2026
Sprint Goal	Implement course announcements, assignment creation with configurable deadlines and late policies, file-based submission tracking, and the student to-do list.
Modules Developed	Announcements & Materials | Assignments & Submissions | To-Do List & Deadlines

2. Sprint 2 Backlog

a) User Stories Selected for Sprint 2

US 10 & US 11 — As an Instructor, I want to post announcements so that I can broadcast important updates or instructions to all enrolled students; as a Student, I want to see an announcement feed for my course so that I never miss critical information from my instructor.

US 14 & US 15 — As an Instructor, I want to create assignments with a title, description, and due date/time so that expectations are clear to students; I also want to define late policies and resubmission rules so that I can enforce grading standards.

US 16, US 17 & US 18 — As a Student, I want to submit my assignments via file upload so that I can turn in my completed work electronically; as an Instructor, I want the system to timestamp all submissions so that I can accurately determine their status; as a Student, I want to see the status of my submissions (submitted, late, missing) so that I know my work has been received.

US 23 & US 24 — As a Student, I want to view a to-do list within my course catalogue section so that I have complete visibility over my progress (submitted vs. missing items); I also want the system to highlight upcoming due dates and late/missed items so that I can effectively manage my deadlines and prioritise my workload.

b) Sub User Stories and Task Breakdowns

US 10 & 11 — Course Announcements

- Design CreateAnnouncement form with title, body text, and inline formatting controls (bold, italic, underline, bullet)
- Support optional file attachments on announcements via system file picker
- Persist announcement record to database linked to course with author ID and server timestamp
- Display announcement feed per course in reverse-chronological order for all enrolled students
- Show author name and post timestamp on each announcement card

US 14 & 15 — Assignment Creation and Policies

- Design CreateAssignment form: title, description, total points, due date (DatePicker), due time (hour/minute spinners)

- Add late submission toggle: 'Allow late turn-ins' vs. 'Hard deadline'
- Persist assignment to database linked to course with late policy flag
- Propagate new assignment to course stream and enrolled students' To-Do lists on creation

US 16, 17 & 18 — Submissions and Tracking

- Implement file upload area within AssignmentDetails view using system FileChooser
- Record server-side timestamp on every submission event
- Auto-tag submission status: Submitted (on time) / Late (after deadline, if policy allows) / Missing (deadline passed, no upload)
- Implement conditional unsubmit / resubmit based on deadline and instructor late policy setting
- Threaded comment area within assignment detail view for both students and instructors

US 23 & 24 — To-Do List

- Implement To-Do view listing all pending assignments across every enrolled course
- Partition list into Active (upcoming) and Missing (deadline passed without submission) sections
- Deadline urgency colour-coding: red (overdue), orange (due within 48 hours), default (normal)
- Auto-refresh list on load; clicking an item opens AssignmentDetails view
- Auto-remove item from active list once submission is recorded

3. Structured Specifications

Spec 1: US 10 & US 11 — Course Announcements

Field	Detail
Actor	Instructor (post); Student (view)
Inputs	Announcement title (required), body text (required), optional file attachments
Processing	Validate title and body are non-empty; insert announcement record with author ID, course ID, and server timestamp; persist any attached files; push to course stream
Outputs	Announcement immediately visible in course feed for all enrolled students, ordered newest-first
Preconditions	Instructor is logged in and owns the course
Postconditions	Announcement record persisted; all enrolled students can view it
Exceptions	Empty title or body → inline validation error; form does not submit

Spec 2: US 14 & US 15 — Assignment Creation and Policies

Field	Detail
Actor	Instructor
Inputs	Title (required), description (optional), total points (required, numeric), due date (DatePicker, required), due time (hour/minute spinners, required), late submission toggle
Processing	Validate title, points, and deadline are non-empty; combine date and time into a LocalDateTime; insert assignment record linked to course with late policy flag

Outputs	Assignment appears in course stream and on all enrolled students' To-Do lists
Preconditions	Instructor is logged in; course exists
Postconditions	Assignment record persisted with deadline and late policy; visible to students
Exceptions	Missing title → validation error; missing deadline → validation error; non-numeric points → validation error

Spec 3: US 16, 17 & 18 — Assignment Submissions and Tracking

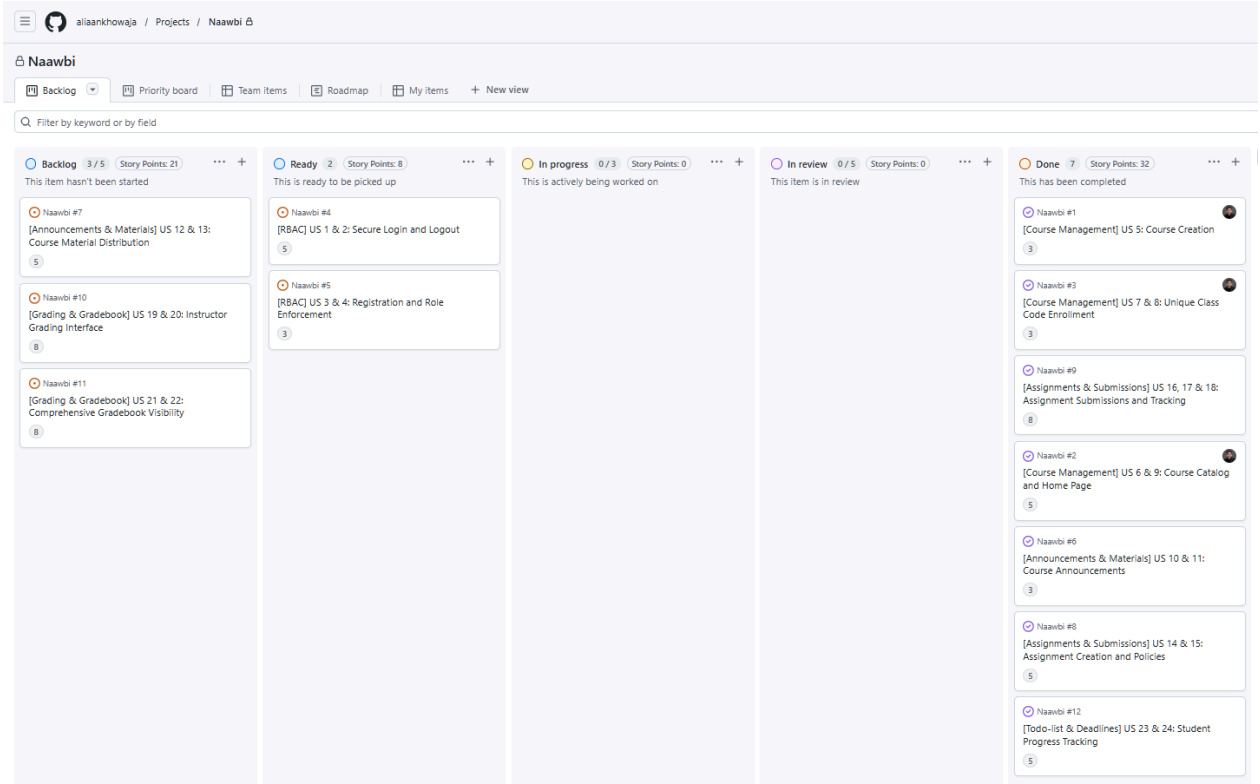
Field	Detail
Actor	Student (submit); Instructor (view status)
Inputs	File selected via system FileChooser
Processing	Record server-side timestamp; compare against assignment deadline; evaluate late policy flag; set status: Submitted (before deadline), Late (after deadline if policy permits), Missing (deadline passed, no upload)
Outputs	Submission record stored; status badge updated in AssignmentDetails view for both student and instructor
Preconditions	Student is enrolled in the course; assignment is accessible and has a set deadline
Postconditions	Submission record created or updated with file path and timestamp; status visible to student and instructor
Exceptions	No file selected → submit button remains inactive until a file is chosen via the file picker; late submission when policy disallows → submission is blocked and the student is notified

Spec 4: US 23 & US 24 — Student To-Do List and Deadline Awareness

Field	Detail
Actor	Student
Inputs	System reads all assignments across enrolled courses at page load
Processing	Fetch all assignments for enrolled courses; filter out submitted items; classify remaining as Active (deadline in future) or Missing (deadline passed without submission); compute urgency relative to current time
Outputs	To-Do view showing Active and Missing sections; items colour-coded red (overdue), orange (due ≤ 48 hours), default (more than 48 hours); clicking any item opens AssignmentDetails
Preconditions	Student is logged in and enrolled in at least one course
Postconditions	Student sees an accurate, up-to-date workload snapshot; submitted items are absent from the list
Exceptions	No pending assignments → empty state message displayed

4. Scrum Board (Github Projects Snapshots)

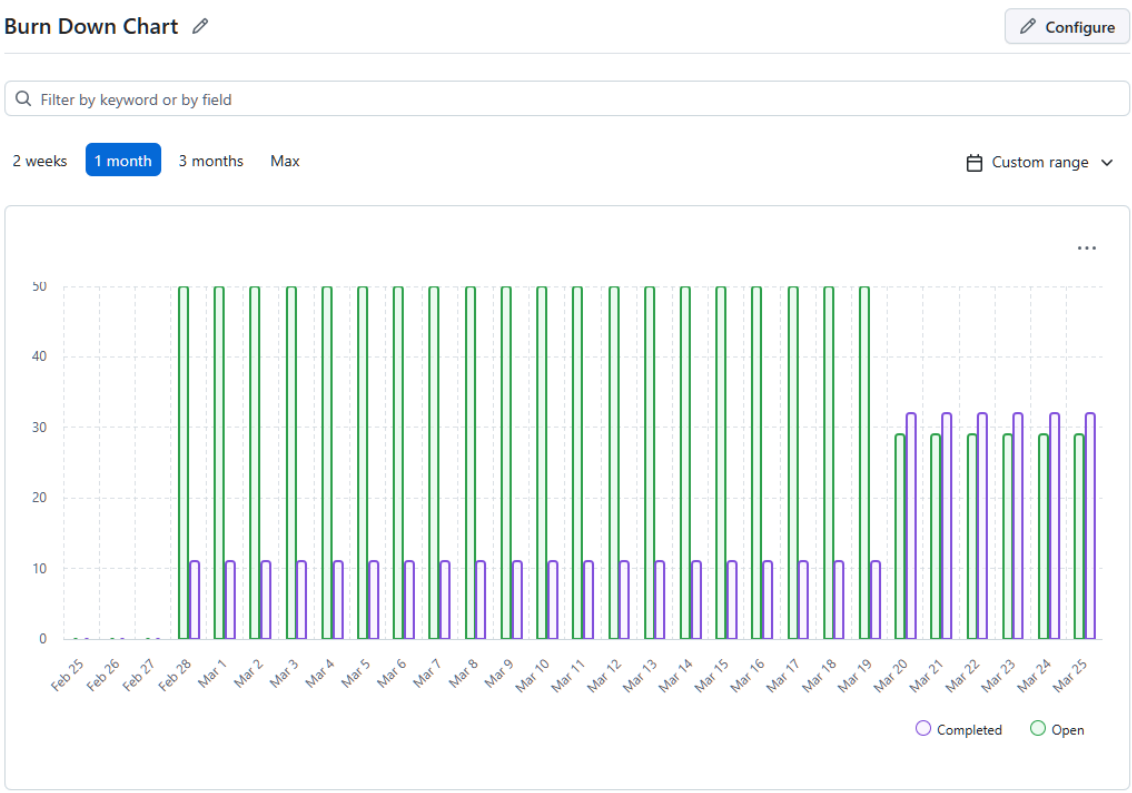
Three snapshots were captured from the Github Projects board at key stages of Sprint 2. The board tracks story progress across Backlog, Ready, In Progress, In Review, and Done columns.



End of Sprint 2: all planned stories complete

5. Sprint 2 Burn-Down Chart

The burn-down chart below compares planned story points (21 SP total) against completed story points across the sprint duration (1 March – 25 March 2026).



Sprint 2 Burn-Down Chart — planned vs. completed story points

PART C — Deliverable 2 Document

1. Introduction

Context

Naawbi is a university-level Learning Management System (LMS) being developed iteratively across multiple sprints by Team Ibwaan for CS3009 Software Engineering (Spring 2026). This document covers Deliverable 2, which represents the output of Sprint 2.

Progress from Sprint 1 to Sprint 2

Sprint 1 established the structural foundation of the platform: course creation, the course catalogue, and the unique-code enrollment workflow. Sprint 2 extends this foundation into the core learning workflow. Instructors can now publish announcements, create assignments with configurable deadlines and late policies, and students can submit work electronically with automatic status tracking. A student-facing To-Do list aggregates pending work across all enrolled courses with urgency indicators, providing proactive deadline awareness.

Problem Statement

The problem remains unchanged from Deliverable 1: fragmented academic tooling forces students to track deadlines, submissions, and grades across multiple disconnected systems. Sprint 2 directly addresses the submission and communication gap — the previously built course structure now has live content flowing through it.

2. Project Vision

Build a clean, reliable, course-centric LMS that instructors can use to manage learning materials and evaluations, and that students can use to track progress and performance.

Sprint 1 delivered the course structure. Sprint 2 delivers the content and submission layer. Future sprints will close the loop with authentication, role enforcement, material distribution, and grading.

Key differentiators:

- Submission tracking — timestamped file submissions with automatic status tagging (Submitted / Late / Missing).
- Deadline awareness — urgency-coloured To-Do list providing students a single view of all pending work.
- Grading pipeline — rubric-driven grading and gradebook (planned for a future sprint).

3. Intended Use of the System

Stakeholder	Primary Need	Sprint 2 Impact
Student	Single place to track work, deadlines, and grades	Can now submit assignments, view submission status, and see a unified To-Do list across all courses

Instructor	Unified workflow for publishing content and collecting work	Can now post announcements and create assignments with deadlines and late policies

4. Features and Overall Functionality

Sprint 1 Features — Completed in Deliverable 1

Feature	User Stories	Status
Course Creation	US 5	✓ Done
Course Catalogue and Home Page	US 6, US 9	✓ Done
Unique Class Code Enrollment	US 7, US 8	✓ Done

Sprint 2 Features — Completed in Deliverable 2

Feature	User Stories	Description
Course Announcements	US 10, US 11	Instructors post announcements; students view a real-time course feed
Assignment Creation and Policies	US 14, US 15	Instructors create assignments with deadlines, point values, and late submission rules
Assignment Submissions and Tracking	US 16, US 17, US 18	Students submit files; system timestamps and tags status automatically
To-Do List and Deadline Awareness	US 23, US 24	Student-facing aggregated pending work list with urgency colour-coding

How Sprint 2 Improved System Capability

Sprint 1 gave users a place to be. Sprint 2 gives them something to do. The course structure is now populated with content (announcements), actionable tasks (assignments), and a feedback mechanism (submission status). The To-Do list converts the system from a passive catalogue into an active workload manager, giving students proactive visibility into pending work across all their enrolled courses.

5. User Stories (Sprint 2)

5.1 Announcements & Materials

US 10 — As an Instructor, I want to post announcements so that I can broadcast important updates or instructions to all enrolled students.

US 11 — As a Student, I want to see an announcement feed for my course so that I never miss critical information from my instructor.

5.2 Assignments & Submissions

US 14 — As an Instructor, I want to create assignments with a title, description, and due date/time so that expectations are clear to students.

US 15 — As an Instructor, I want to define late policies and resubmission rules so that I can enforce grading standards.

US 16 — As a Student, I want to submit my assignments via file upload so that I can turn in my completed work electronically.

US 17 — As an Instructor, I want the system to timestamp all submissions so that I can accurately determine their status (submitted, late, missing).

US 18 — As a Student, I want to see the status of my submissions (submitted on time, late, missing) so that I know my work has been received.

5.3 To-Do List & Deadlines

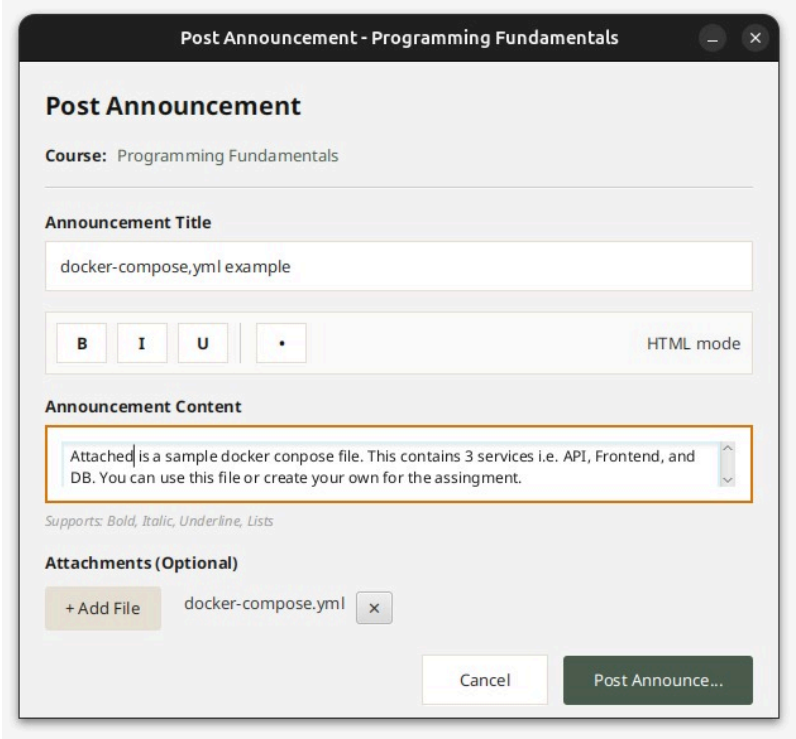
US 23 — As a Student, I want to view a to-do list within my course catalogue section so that I have complete visibility over my progress (submitted vs. missing items).

US 24 — As a Student, I want the system to highlight upcoming due dates and late/missed items so that I can effectively manage my deadlines and prioritise my workload.

6. Non-Functional Requirements (NFR) — Updated

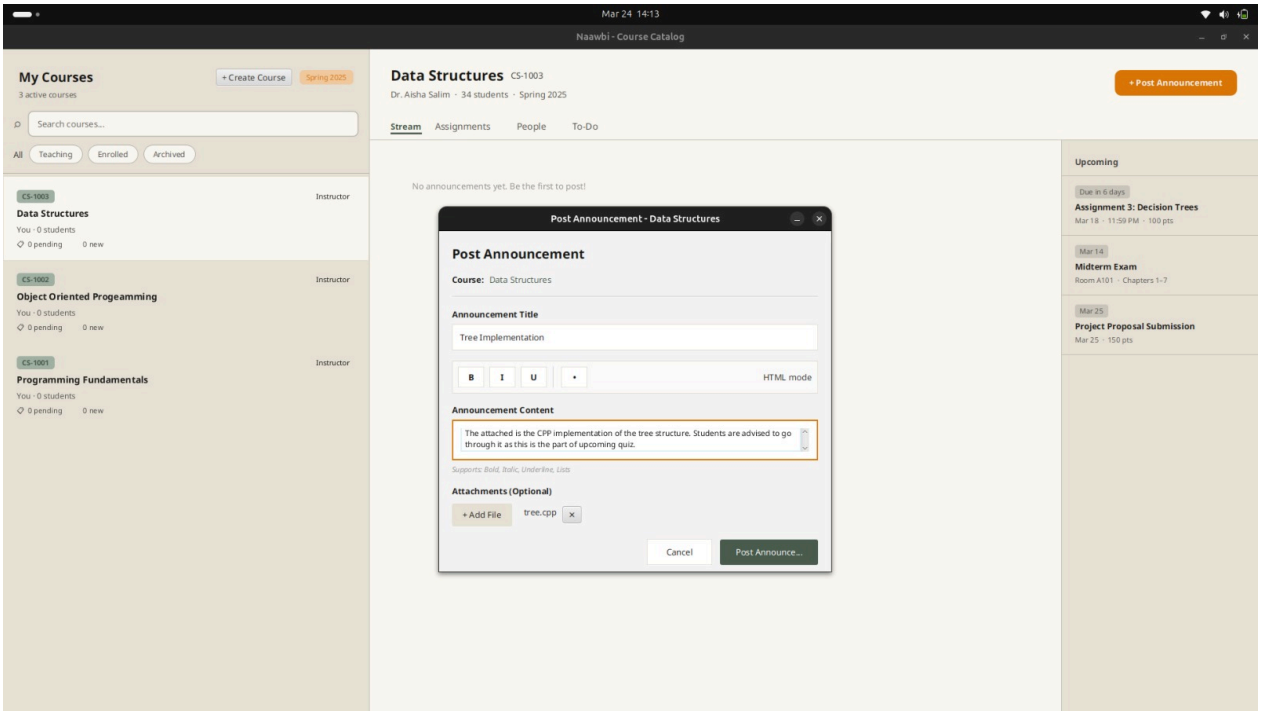
NFR Category	Requirement
Performance	Primary pages (course home, assignment list, To-Do list) load within 2–3 seconds under normal class sizes (~200 students per course) on typical lab hardware. File uploads complete within a reasonable time for typical submission sizes (documents and PDFs up to 10 MB).
Security	Students can only view their own submission files and status. Instructors can only access courses they own. User inputs (announcement text, assignment descriptions) are validated and sanitised. File uploads are restricted to expected types to prevent malicious uploads.
Reliability	Submissions are not lost once acknowledged by the system. Deadline-based status (Submitted / Late / Missing) is computed consistently using server time, not client time. Errors are surfaced with actionable messages rather than silent failures.
Usability	Consistent navigation and layout across all Sprint 2 modules. Status badges (Submitted / Late / Missing) use both colour and text so they are not reliant on colour alone. To-Do list urgency colours (red, orange) are consistent with the status badge colour palette.
Maintainability	Layered architecture (controller / model / view) maintained across all new modules. Business logic (deadline evaluation, status tagging) is encapsulated in model classes, not in UI controllers.
Scalability	To-Do list and submission tracking queries are designed to handle multiple courses and hundreds of assignments without requiring denormalised data structures.
Auditability	Submission events (file upload, unsubmit, resubmit) are logged with server timestamps. Announcement creation is logged with author and timestamp. These records are preserved and not overwritten on resubmission.

7. Iteration 2 Implementation Screenshots



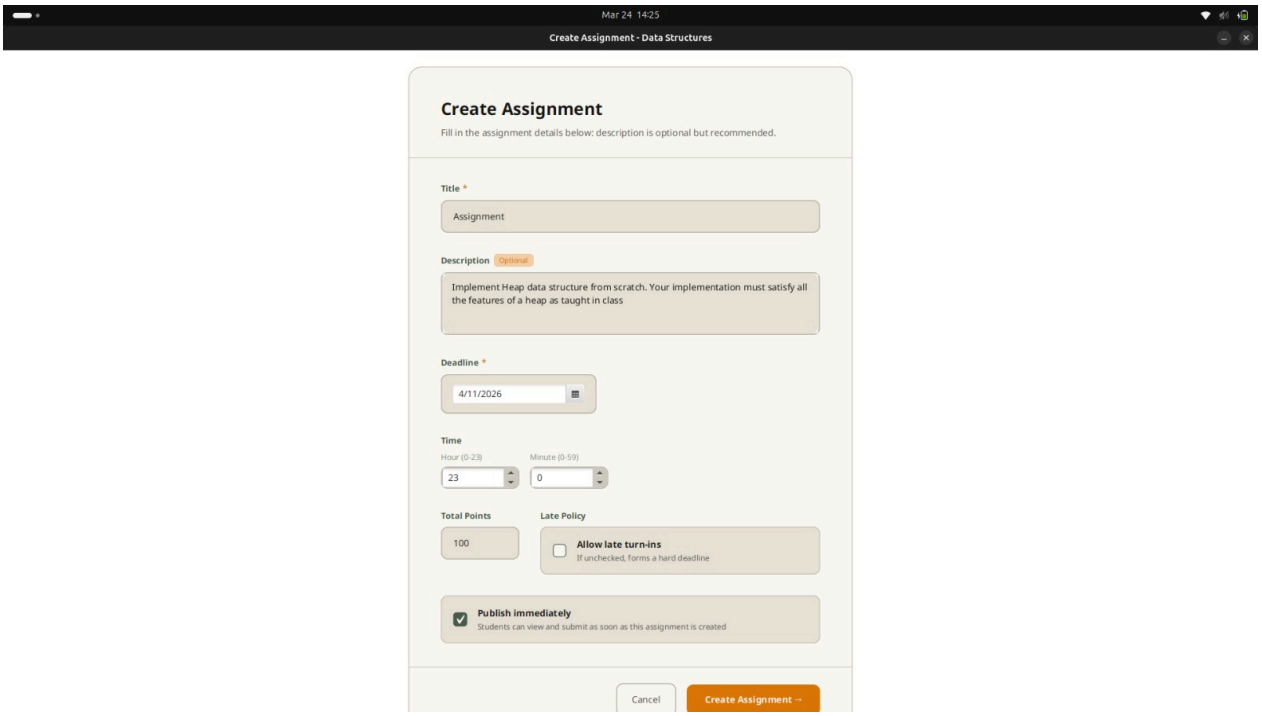
Screenshot 1 — Create Announcement view

Create Announcement — instructor form with title, body, formatting controls (bold/italic/underline/bullet), and optional file attachment.



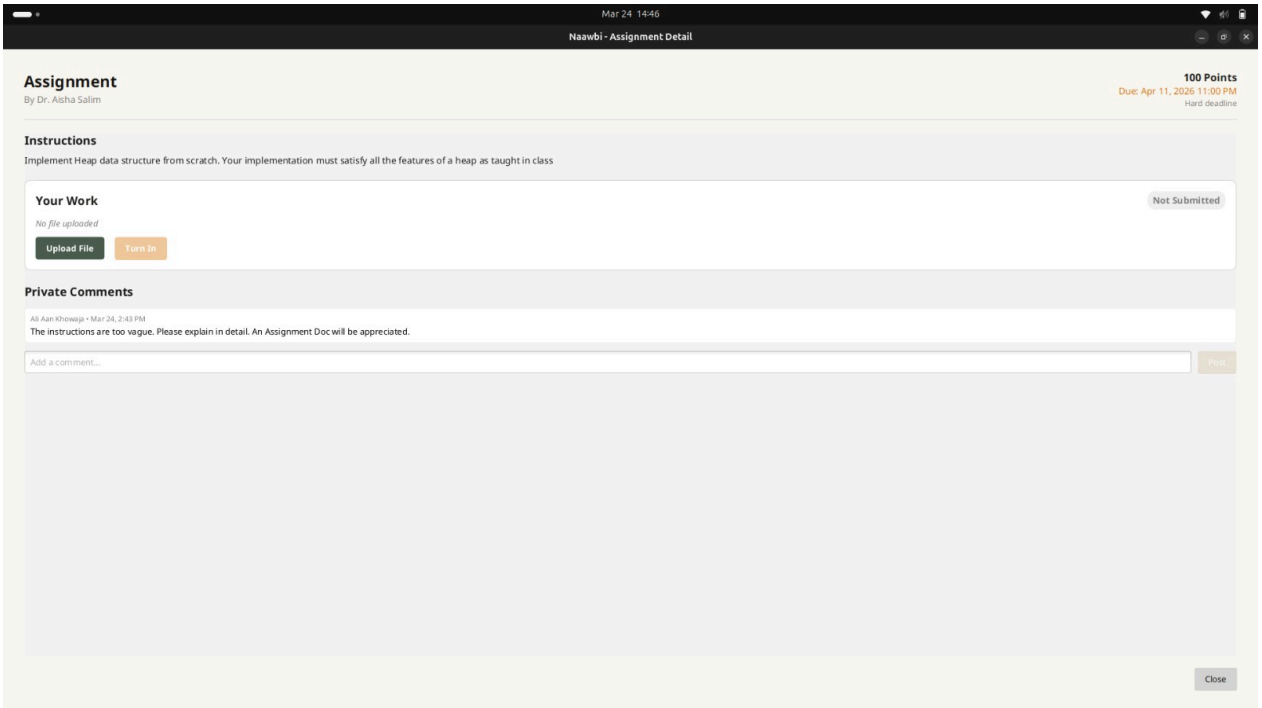
Screenshot 2 — Course stream

Course Stream — reverse-chronological feed displaying announcements with author name and timestamp.



Screenshot 3 — Create Assignment dialog

Create Assignment — instructor form with title, description, total points, date picker, time spinners, and late submission toggle.



Screenshot 4 — Assignment Details with submission area

Assignment Details — student view showing assignment metadata, file upload area, submission status badge (Submitted / Late / Missing), and threaded comment section.

8. Work Division for Sprint 2

Member	Sprint 2 Responsibilities
Ibraheem Farooq (PM / Scrum Master / Developer)	Sprint 2 planning and tracking. Maintains Github Projects board state across sprint stages. Final review and packaging of deliverable artefacts. Branch/PR oversight and release readiness. Contributed to backend development for course and announcement modules.
Wajih-Ur-Raza Asif (Requirements Architect / Tester / Developer)	Sprint 2 user stories, sub-story breakdowns, and structured specifications. Consistency verification between acceptance criteria and implemented behaviour. Deliverable document authoring and completeness review. Contributed to development and testing of submission tracking and To-Do list logic.
Ali Aan Khowaja (Developer / UI Designer)	Lead implementation of all Sprint 2 features: Announcements, Assignment Creation, AssignmentDetails with submission flow, and To-Do List. UI design and screen layout for all new views. Produces Sprint 2 implementation screenshots and demo-ready flows.